

Yajnaseni Dutta

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Universiteit Leiden – Mathematisch Instituut
Gorlaeus Gebouw, Rm. BW 2.34, Leiden, NL.

EMPLOYMENT AND AFFILIATIONS

- Assistant Professor (UD)** Sep. 2023–present
Universiteit Leiden, Mathematical Institute (tenured Nov. 2024)
Responsibility: Research:Teaching:Admin = 2:2:1
- Postdoctoral Researcher** Sep. 2022–Aug. 2023
Rheinische Friedrich-Wilhelms-Universität Bonn, Complex Geometry Group of Prof. Daniel Huybrechts
- Hausdorff Postdoctoral Researcher** Sep. 2019–Aug. 2022
Hausdorff Center for Mathematics Bonn, Complex Geometry Group of Prof. Daniel Huybrechts

SELECTED RESEARCH VISITS

- Oct. 2023 Visitor (ten days), **Universität Bonn**. Host: Prof. Daniel Huybrechts
- Oct. 2019–Dec. 2019 Visiting Scholar (three months), **Katholieke Universiteit Leuven**. Host: Prof. Nero Budur

EDUCATION AND QUALIFICATIONS

- Oct. 2023–Oct. 2025 **University Teaching Qualifications (BKO)**, Universiteit Leiden
- Sep. 2014–Aug. 2019 **Ph.D. Mathematics** at Northwestern University, USA. Advisor: Prof. Mihnea Popa
- Aug. 2013–Aug. 2014 **Ph.D. candidate Mathematics** at University of Illinois at Chicago, USA. Advisor: Prof. Mihnea Popa
- Aug. 2011–Jul. 2013 **M.Sc., Mathematics** at Chennai Mathematical Institute, India
- Aug. 2008–Jul. 2011 **B.Sc., Mathematics and Computer Science** at Chennai Mathematical Institute, India

RESEARCH STATEMENT

I have expertise in a wide range of subfields in algebraic geometry and am able to successfully export ideas and techniques from one field to another. My PhD thesis includes, among other things, a Hodge module theoretic approach to certain questions in birational geometry and topology of algebraic maps. During my postdoc years in Bonn I shifted my focus to hyperkähler geometries, especially Lagrangian fibrations, where I now combine techniques from moduli of sheaves, derived categories, D-modules and Hodge theory. Following this trajectory, one of my latest directions employs techniques from derived de Rham cohomology, and motivic cohomology.

LIST OF PUBLICATIONS/PREPRINTS**

**In algebraic geometry, like in several other pure mathematics disciplines, it generally takes longer to prepare and publish new results. The authorship is always alphabetical.

9. **Relative compactified Prym and Picard fibrations associated to very good cubic fourfolds.** jt. with L. Marquand, Proc. Amer. Math. Soc., Vol. 154, pp. 25-34(2026), DOI: [10.1090/proc/17417](https://doi.org/10.1090/proc/17417)
8. **Twists of intermediate Jacobian fibrations.** jt. with D. Mattei, and E. Shinder, preprint available at [arXiv:2411.01953v1](https://arxiv.org/abs/2411.01953v1)

7. **Some density results for hyperkähler manifolds.** jt. with E. Izadi, L. Kamenova, and L. Marquand, Math. Res. Lett., Vol. 32, No. 5, pp. 1465-1479 (2025) [10.4310/MRL.251214175204](https://doi.org/10.4310/MRL.251214175204)
6. **On symplectic birational self-maps of projective hyperkähler manifolds of $K3^{[n]}$ -type.** jt. with D. Mattei and Y. Prieto-Montañez, International Mathematics Research Notices, vol. 5 (2024) [doi:10.1093/imrn/rnae112](https://doi.org/10.1093/imrn/rnae112)
5. **Maximal variation of curves on $K3$ surfaces** jt. with D. Huybrechts, Tunisian J. Math. vol. 4, 2022, no. 3 [doi:10.2140/tunis.2022.4.443](https://doi.org/10.2140/tunis.2022.4.443)
4. **Generic Vanishing, 1-forms, and Topology of Albanese Maps** jt. with F. Hao and Y. Liu. Math. Zeitschrift. Volume 306, no. 56, (2024) [doi:10.1007/s00209-024-03438-3](https://doi.org/10.1007/s00209-024-03438-3)
3. **Vanishing for Hodge ideals on toric varieties** Mathematische Nachrichten. vol. 293, 2020. [doi:10.1002/mana.201800363](https://doi.org/10.1002/mana.201800363)
2. **Effective generation and twisted weak positivity of direct images** jt. with T. Murayama. Algebra and Number theory. vol. 13, no. 2, 2019. [doi:10.2140/ant.2019.13.425](https://doi.org/10.2140/ant.2019.13.425)
1. **On the effective freeness of the direct images of pluricanonical bundles** Annales de l'Institut Fourier, vol. 70, no. 4, 2020. [doi:10.5802/aif.3351](https://doi.org/10.5802/aif.3351)

AWARDS, FELLOWSHIPS AND RECOGNITION

Sep. 2023	Starter budget for hiring a PhD student (€240,000), Universiteit Leiden
May 2019	Best Thesis in Mathematics , Northwestern University
May 2018	Gelfand Award , Northwestern University (awarded to the Ph.D. student who has contributed most to the department)
2013–2014	University Fellowship , University of Illinois at Chicago (nominated by the Maths Department among the top 15% of newly recruited graduate students, receiving a full year's stipend and tuition waiver)
2008–2011	Scholarship for Higher Education , Department of Science and Technology, Govt. of India (a component of the programme "Innovation in Science Pursuit for Inspired Research (INSPIRE)")
2007	Qualified regional mathematical olympiad , West Bengal (unique track for being part of the IMO-team, India and is held across regions throughout India)

THESIS SUPERVISION

PhD	Céline Fietz (with Prof. E. Shinder), Universiteit Leiden, 2023 –
Master's	Matthijs Buijsen , Moduli of Sheaves (tentative), Universiteit Leiden, Sep 2025– Jorge Macein Sanz (with Dr. E. Brakkee), Elliptic Fibrations on the Fermat Quartic, Universiteit Leiden, Mar 2025 - Jan 2026 Céline Fietz (with Prof. E. Shinder), Categorical Resolutions of A2 Singularities, Universität Bonn, Oct. 2022–Oct. 2023
Bachelor's	Lucio Verrijzer , The Morita Equivalence and (Graded) Brauer Groups, Universiteit Leiden, Feb.–Jun. 2024

TEACHING

Spring'27	Not teaching
Fall'26	Advanced Algebraic Geometry: Abelian Varieties , instructor, national mastermath program and Algebraic Curves , Universiteit Leiden
Current	Mathematics and Networks , instructor, Leiden University College

Netherlands	Algebraic Curves , Leiden, Mathematical Reasoning , LUC, Representation Theory , Leiden, Advanced Algebraic Geometry: Moduli of sheaves , national master-math program
Bonn	Algebraic Geometry I, II , coordinator teaching assistants, Hauptseminar Algebra: Klassische Algebraische Geometrie , instructor, Selected Topics in AG: Moduli of Vector Bundles , instructor, Graduate Seminar: Algebraic Curves and Abelian Varieties , instructor
Northwestern	Calculus (all levels), Linear Algebra , Foundations of Higher Math , Intro to Topology , Graph Theory , teaching assistant, Differential Topology , Algebraic Geometry , grader

SELECTED WORKSHOPS (2020–)

Jul. 2022, 26	Algebraic Geometry: Moduli Spaces, Birational Geometry and Derived Aspects , Mathematisches Forschungsinstitut Oberwolfach
Apr. 2026	Complex Analysis , Mathematisches Forschungsinstitut Oberwolfach
Sep. 2023	Geometry of HyperKähler Varieties , IASM Hangzhou
Aug. 2023	Women in Mathematical Physics II , Banff, Canada (participated online)
Apr. 2023	Arbeitsgemeinschaft: Twistor D-Modules and the Decomposition Theorem , Mathematisches Forschungsinstitut Oberwolfach

SELECTED TALKS (2020–)

Selected Conference Talks

2026	Simons Workshop on Higher Dimensional Varieties , Erdős Center, Budapest, October LMS symposium: Rationality problems, Cremona groups, and motivic invariants , Sheffield, July
2025	Conference in Algebraic Geometry via Eurostar , Leiden, Netherlands, September Japanese European Symposium on Symplectic Varieties and Moduli Spaces , Netherlands, September
2024	D-modules, local systems and applications , CRM Montreal, September GQT section: Netherlands Mathematical Congress , Lunteren, April
2023	Belgian–Dutch Algebraic Geometry Meeting , Leiden, October Geometry of HyperKähler Varieties , Hangzhou, September Derived categories, Hodge theory and Singularities , Sheffield, June
2022	Algebraic Geometry section: DMV-Jahrestagung , FU Berlin, September D-Modules: Applications to Algebraic Geometry , Arithmetic and Mirror Symmetry, CIRM (Luminy), April
2021	Singularities in the Midwest Workshop , Wisconsin (Online), May
2020	AMS Fall Eastern Sectional, "Algebraic Singularities in Arbitrary Characteristic" , (Online), October

Lecture Series

2026	Intermediate Jacobians – lecture series (5 hours), Swiss-French Algebraic Geometry Workshop
2025	D-modules in Lagrangian fibrations – lecture series (3 hours), D-modules in Bochum

General Audience Talks

2026	Abel Prize Evening , Royal Dutch Academy of Arts and Sciences (KNAW), February https://www.youtube.com/watch?v=CPKoAPxwRWA
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Selected Seminar Talks

(2025) Groningen, (2024) Universiteit van Amsterdam, Harvard, Johns Hopkins, Strasbourg, Princeton, (2023) USTC Hefei, Essen, (2022) IMSC Chennai, Columbia (2021-online) Stony Brook, Princeton, San Diego, Edinburgh, Bonn (2020-online) Frankfurt, TIFR Mumbai, Hannover, CMI Chennai

ORGANISATION AND PROFESSIONAL SERVICES

Service in Research Community

Leiden organised **workshop: K3 surfaces & friends**: Brauer groups and moduli, Lorentz Center, Leiden, July 2025
organised **Intercity Geometry Seminar**, Amsterdam, Leiden, Utrecht, Antwerp (Dutch intercity study seminar), 2024–
organised: **Hodge Theory in Leiden**, a conference in honour of Prof. Christ Peters' 70th birthday
organised mini-symposium: **Around Hyperkählers and K3s**, Leiden, December 2023
Journals **Referee and quick opinion** Inventiones Mathematicae, Crelle, Math. Zeitschrift, Math. Nachrichten, Forum of Math (Sigma), PRIMS Kyoto, Chinese Annals of Mathematics
Grant refereed OPUS proposal for Narodowe Centrum Nauki (Poland)
Reviewing zbMATH
PhD Dissertation referee for D. van Bree (Utrecht), D. Leijnse (UvA)

Outreach and Diversity

GROW Founding organiser of **Graduate Research Opportunities for Women (GROW)**, a yearly conference aimed at female bachelor students for promoting math academia as a career, 2021-23, Bonn
Founding member of the **steering committee** ensuring continuity and unanimity of the GROW conferences across Europe, 2025–
Leiden Initiated the formation of **Women* student chapter** at the mathematical institute of Leiden. Led by five female students the group organises academic and team-building events open to all students irrespective of gender.
NUAWM **panel member for graduate school applications for NUAWM** (Northwestern University Association for Women in Mathematics), an undergraduate club.

Selected Organisational Roles

Leiden **Study Advisor** for masters students, Algebra, geometry and number theory
Consultation (Overleg) team, an advisory board to the management
Bonn **Recruitment committee member** for W3 professorship in mathematics
Northwestern **panel member for TA seminar** for 2nd year graduate students

MISCELLANEOUS INFORMATION

Languages Fluent in English, Bengali (native), Hindi. General professional proficiency (Goethe-Zertifikat B2) in German. Limited Working Proficiency in Dutch.